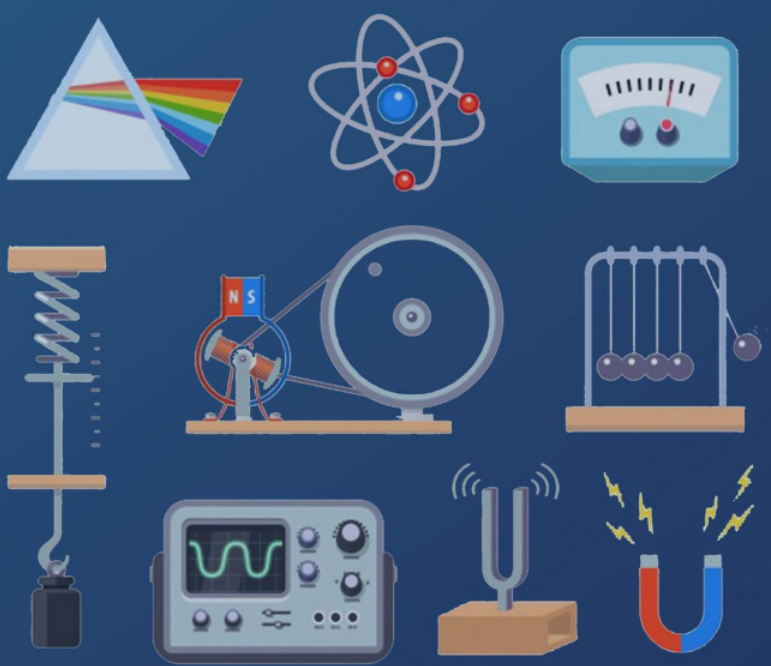
PHYSICS

Units And Dimensions



SHEET

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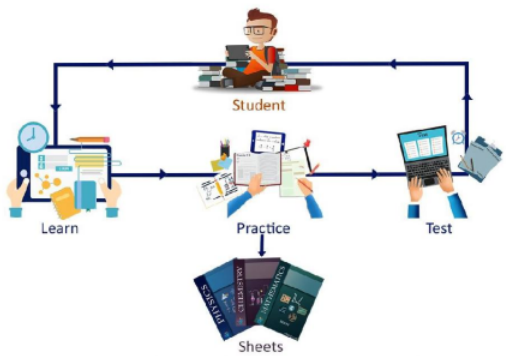
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topic	Ex - 1	Ex - 1A	Ex - 2	Ex - 2A
1	Units and Dimensions	1, 2, 3, 4, 5	1,2,6,7		
2	Applications of Dimensions	6, 7, 8, 9, 10	3,4,8	1, 2, 3, 4	1,2,3,6
3	Basic Maths	11, 12	5	5	4, 5

Exercise - 1 — Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Units and Dimensions

Q1. Which of the following system of units is not based on units of mass, length and time alone.

(A) SI (B) MKS
(C) FPS (D) CGS

Q2. Dimensions of frequency are:-

(A) $M^0L^{-1}T^0$ (B) $M^0L^0T^{-1}$
(C) M^0L^0T (D) MT^{-2}

Q3. Which is dimensionless :-

(A) $\frac{\text{Force}}{\text{Acceleration}}$ (B) $\frac{\text{Velocity}}{\text{Acceleration}}$
(C) $\frac{\text{Volume}}{\text{Area}}$ (D) $\frac{\text{Energy}}{\text{Work}}$

Q4. The M.K.S. units of coefficient of viscosity is :-

(A) kg m^{-1} s^{-1} (B) kg m s^{-2}
(C) kg m^2 s^{-1} (D) kg^{-1} $m^{-1}s^2$

Q5. The dimensional formula for which of the following pair is same.

(A) Force and momentum
(B) Kinetic energy and work
(C) strain and pressure
(D) momentum and angular momentum

Applications of Dimensions

Q6. Which of the following functions of A and B may be performed if A and B possess different dimensions?

(A) $\frac{A}{B}$ (B) A + B
(C) A – B (D) None

Q7. Density of wood is 0.5 g/ cc in the CGS system of units. The corresponding value in MKS units is-

(A) 500 (B) 5
(C) 0.5 (D) 5000

Q8. The time dependence of a physical quantity P is given by $P = P_0\, e^{-\alpha t^2}$, where α is a constant and t is time. The constant α

(A) Is dimensionless
(B) Has dimensions T^{-2}
(C) Has dimensions of P
(D) Has dimensions T^2

Q9. Which of the following can be a set of fundamental quantities.

(A) length, velocity, time
(B) momentum, mass, velocity
(C) force, mass, velocity
(D) momentum, time, frequency

Q10. If area (A), velocity (v), and density (ρ) are base units, then the dimensional formula of force can be represented as -

(A) $Av\rho$ (B) $Av^2\rho$
(C) $Av\rho^2$ (D) $A^2v\rho$

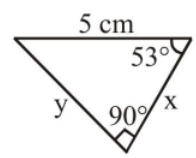
Basic Maths

Q11. Value of $(1.002)^3$ = ...

(A) 1.008 (B) 1.004
(C) 1.006 (D) 1.005

Q12. Find x, y and perimeter of the triangle

(A) 3 cm, 4 cm, 10 cm
(B) 4 cm, 3 cm, 12 cm
(C) 3 cm, 4 cm, 12cm
(D) 5 cm, 4 cm, 15 cm



ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1.

Column I	Column II	Column III
Physical quantity	MKS system	CGS system
(I) Length	(A) newton	(P) erg
(II) Mass	(B) m	(Q) dyne
(III) Force	(C) kg	(R) erg/s
(IV) Power	(D) joule	(S) g
(V) Work	(E) watt	(T) cm

Which of the following options have matching columns :

(A) I–B–T (B) V–C–S

(C) III-A–Q (D) IV–E–R

Q2. Choose the INCORRECT statement(s) :-

(A) A dimensionless quantity must be unitless

(B) A dimensionless quantity may have a unit

(C) A unitless quantity may have non zero dimension

(D) Two quantities cannot have same dimension

Q3. Which of the following is dimensionally correct formula :-

(A) $s = ut + \frac{1}{2}at^2$

(B) $F = ma$

(C) kinetic energy $= mv^2$

(D) $vt = u - a$

Q4. Choose the correct statement(s) :-

(A) A dimensionally correct equation may be correct

(B) A dimensionally correct equation may be incorrect

(C) A dimensionally incorrect equation may be correct

(D) A dimensionally incorrect equation must be incorrect

Q5. Find the correct option(s) (here $2^c = 2$ radian):-

(A) $\sin 2° = 2$ (B) $\sin 2° = 2 \times \frac{\pi}{180}$

(C) $\sin 2^c = 2$ (D) $\sin 30° = 0.5$

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q6.

Column I	Column II
(A) Coefficient of viscosity	(P) Dimensionless
(B) Strain	(Q) Unitless
(C) Angle	(R) $ML^{-1}T^{-1}$
(D) Stress	(S) $ML^{-1}T^{-2}$

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

Q7. If $M^a L^b T^c$ is the dimensional formula of momentum and $M^x L^y T^z$ dimensional formula of energy find the value of ax + by – cz = ?

ASSERTION AND REASON

Q8. **Statement-1 :** Method of dimensions cannot tell whether an equation is correct.

Statement-2 : A dimensionally incorrect equation may be correct.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. If velocity $v = \frac{A}{Bt^2}\sin(Bx^2 - Ct^3)$ is correct formula, where t is time and x is length then dimensions of A will be:-

(A) L^1T^1 (B) $L^{-1}T^1$

(C) L^1T^{-1} (D) None

Q2. If force, acceleration and time are taken as fundamental quantities, then the dimensions of length will be:

(A) FT^2 (B) $F^{-1}A^2T^{-1}$

(C) FA^2T (D) AT^2

Q3. If the unit of length be doubled then the numerical value of the universal gravitation constant G will become (with respect to present value)

(A) Double (B) Half

(C) 8 times (D) 1/8 times

Q4. If in a system, the force of attraction between two point masses of 1 kg each situated 1 km apart is taken as a unit of force and is called notwen (newton written in reverse order) & if $G = 6.67\times 10^{-11}\,N\text{–}m^2\,kg^{-2}$ in SI units then which of the following is true?

(A) 1 notwen = 6.67×10^{-11} newton

(B) 1 newton = 6.67×10^{-17} notwen

(C) 1 notwen = 6.67×10^{-17} newton

(D) 1 newton = 6.67×10^{-12} notwen

Q5. The angle subtended by the planet's diameter at a point on the earth is about 1°. Use this and the fact that the planet is about 1800000 km away to find the approximate diameter of the planet.



(A) 3140 km (B) 31400 km

(C) 314000 km (D) 3140000 km

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph for Question No. 1 to 2

Dimensional methods provide three major advantage in verification, derivation and to change the system of units. Any emperical formula that is derived based on this method has to be verified and proportionality constants found by experimental means. The presence or absence of certain factors (numerical constants of 1, 2, π etc) cannot be identified by this method. So every dimensionally correct relation cannot be taken as perfectly correct.

Q1. Energy (E) of a oscillating particle is dependent on mass M, frequency 'f' and amplitude 'A' of oscillation. Then energy 'E' is proportional to -

(A) $\frac{Mf}{A^2}$ (B) Mf^2A^2

(C) Mf^2A^{-2} (D) MfA^{-2}

Q2. Which of the following relations is dimensionally incorrect ?

(A) $g = \frac{4\pi^2\ell}{T^2}$ where g is acceleration due to gravity, ℓ is length, T is time period

(B) $v = \sqrt{\frac{2k}{m}}$; m → mass ; k → kinetic energy ; v → speed

(C) $\frac{F}{m^2} = \frac{G}{r^2}$; F → force, m → mass ; G → gravitational constant ; r → distance

(D) $v^2 = u^2 + 2as^2$; v → final velocity ; u → initial velocity ; a → acceleration ; s → displacement

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q3. In a new system of units known as RMP, length is measured in 'retem', mass is measured in 'marg' and time is measured in 'pal'.

100 retem = 1.0 meter

1.0 marg = 10^{-3} kilogram

10 pal = 1.0 second

In the given table some unit conversion factors are given. Suggest suitable match.

Column - I	Column - II
(A) One SI unit of force	(P) 10^2 units of RMP
(B) One SI unit of potential energy	(Q) 10^3 units of RMP
(C) One SI unit of power	(R) 10^4 units of RMP
(D) One SI unit of momentum	(S) 10^5 units of RMP
	(T) 10^6 units of RMP

NUMERICAL TYPE

*The answer to each question is a **NUMERICAL VALUE**. For each question, find the correct numerical value (in decimal notation, truncated/ rounded-off to the second decimal place; e.g. 6.25, 7.00, -0.33, -.30, 30.27, -127.30)*

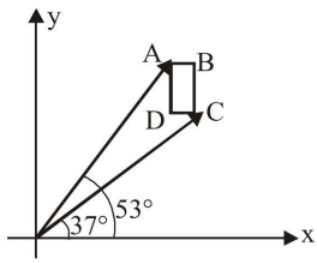
Q4. Use the approximation

$(1+x)^n \approx 1+nx, |x| << 1$,

to find approximate value for -

(A) $\sqrt{99}$ (B) $\frac{1}{1.01}$

Q5. The farm house shown in figure has rectangular shape and has sides parallel to the chosen x and y axes. The position vector of corner A is 125 m at 53° and corner C is 100 m at 37° from x axis. Find the length of the fencing required in meter.



Q6. A particle of mass m is in a uni–directional potential field where the potential energy of a particle depends on the x–coordinate given by $\phi_x = \phi_0 (1 - \cos ax)$, where '$\phi_0$' and 'a' are constants. Find the physical dimensions of 'a' & ϕ_0.

Answer Key

Ex-1

1	2	3	4	5	6	7	8	9	10	11	12
A	B	D	A	B	A	A	B	C	B	C	C

Ex- 1A

1. A,C,D **2. A,C,D** **3. A,B,C** **4. A, B, D** **5. B, D**

6. (A)-R, (B)-P,Q; (C)-P, (D)-S **7. 1** **8. C**

Ex-2

1	2	3	4	5
B	D	D	C	B

Ex- 2A

1. B **2. D** **3. (A)-Q, (B)-S, (C)-R, (D)-R**

4. (A) 9.95, (B) 0.99 **5. 90 m** **6. L^{-1}, ML^2T^{-2}**

-: Important Notes & Formulas :-